

Yuxin Shen

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EDUCATION

University of Edinburgh, School of Biological Science, Edinburgh, UK

PhD student in Biological Science (with Internship)

Oct 2022 – present

- Thesis Topic: Machine learning for sequence-to-expression modelling and optimization
- Working on computational biology, applying machine learning to the problems in biological systems
- Funded by BBSRC EASTBIO DTP

Imperial College London, Department of Chemical Engineering, London, UK

MSc in Advanced Chemical Engineering

Oct 2020 – Sep 2021

- Master of Science with Distinction (Grade: 76.26/100)
- Core Modules: Modelling of Biological Systems, Advanced Bioprocess Engineering, Advanced Process Operations, Dynamical Systems in Chemical Engineering, Biochemical Engineering, Advanced Process Optimisation
- Master's thesis: Surrogate methods in model-based design of experiments for parameter precision

Fudan University, Department of Chemistry, Shanghai, China

BSc in Chemistry

Sep 2016 – Jul 2020

- Core Modules: Physical Chemistry I, II, III, Organic Chemistry I, II, Inorganic Chemistry, Analytical Chemistry I, II, Introduction to Modern Biological Science, Topics in Organic Chemistry, and related experiments
- Selected for Summer Programme in Chemical Biology at University of Hong Kong

SCIENTIFIC RESEARCH EXPERIENCES

PhD Research

School of Biological Science, University of Edinburgh

Oct 2022 – present

Supervisor: Prof. Diego Oyarzún and Prof. Grzegorz Kudła

Machine learning for sequence-to-expression mapping

- Integrated biological domain knowledge into feature engineering for machine learning models.
- Worked on both supervised learning (including RF, MLP, CNN, GNN and DNA large language models) and unsupervised learning for sequence-to-expression modelling.
- Built a machine learning model for membrane protein expression with high prediction accuracy and explainability validated in wet lab experiments.

Active learning for biological sequence optimization

- Created an active learning pipeline for DNA sequence generation and selection for protein expression optimization.

MSc Research

Department of Chemical Engineering, Imperial College London

Nov 2020 – Sep 2021

Supervisor: Prof. Benoit Chachuat

A comparative study of surrogate methods in model-based design of experiments

- Applied different surrogates (simple neural network and Bayesian neural network) to approximate chemical reaction model sensitivities.
- Conducted model-based design of experiments based on approximate sensitivities.

RESEARCH OUTPUT

1. Shen, Y., Kudla, G. and Oyarzún, D.A., 2025. Improving the generalization of protein expression models with mechanistic sequence information. *Nucleic Acids Research*, 53(3), gkaf020.
2. Shen, Y., Kudla, G. and Oyarzún, D.A., 2025. Optimization of regulatory DNA with active learning. *Computational and Structural Biotechnology Journal*, 27, pp.4384-4392.
3. Shen, Y., Underhill, J., Mulholland, A.J., Oyarzun, D.A. and Curnow, P., 2025. Effective sequence-to-expression prediction for membrane proteins using machine learning and computational protein design. *bioRxiv*, pp.2025-09.
4. Çubuk, H., Plech, M., Aslanzadeh, V., Zikanova, M., Skopova, V., Kmoch, S., Shen, Y., Marsh, J.A. and Kudla, G., 2025. Mechanistic Modelling of Recessive Disease through Allelic Integration of Variant Effects. *bioRxiv*, pp.2025-08.

WORKING & INTERNSHIP EXPERIENCES

Informatics (Machine learning) Intern

Etcmbl, Oxfordshire, UK

May 2025 – Jul 2025

- PIPS (Professional Internships for PhD Students) of BBSRC EASTBIO PhD Programme

- Developed and implemented a pipeline of Large Language Models (LLMs) and biological domain knowledge for antibody and TCR sequence design.
- Designed DNA sequences with improved performance in wet lab validation.

Formulation R&D Technologist

Research and Development, Unilever (China) Shanghai Branch

Jan 2022 – May 2022

- Worked on product formulation development in laundry products for Homecare sector.
- Focused on new functions and sustainable solutions in liquid/concentrated laundry products.

Bioprocess Engineering Intern

Pall Corporation, Danaher Life Science Early Career Program

Sep 2021 – Dec 2021

- Assisted upstream bioprocess design for gene therapy and monoclonal antibody production.
- Designed single-use systems for bioreactors and the filtration processes.

Bioanalysis Intern

Industrial Research Institute, Shanghai Fosun Pharmaceutical (Group) Co., Ltd.

Dec 2019 – Feb 2020

- Assisted the PK/PD bioanalysis of the APIs for the generic drugs.
- Reviewed LC-MS methodology validation reports and bioanalytical reports.

HONORS & AWARDS

BBSRC EASTBIO PhD studentship (4-year full scholarship)

Oct 2022

Graduate Scholarship of Department of Chemistry

Jun 2020

Professional Scholarship of Department of Chemistry

Oct 2018 & Oct 2017

Outstanding Student Scholarship of Fudan University

Oct 2019 & Oct 2018 & Oct 2017

First Prize in Chinese Chemistry Olympiad in Shanghai

Oct 2015

OTHER EXPERIENCES & PROJECTS

Alan Turing Institute Interest Group seminar series: Co-organizer

Sep 2024 – present

- Co-organize the Data-Centric Biological Design & Engineering Interest Group.
- Invite speakers and organize seminars.
- Update the seminar website, advertise the seminar series.

BBSRC EASTBIO DTP Student Representative

Oct 2022 – Sep 2024

EASTBIO student representative for College of Science and Engineering, University of Edinburgh

- Organized panel and poster sessions in Annual EASTBIO Symposiums.
- Worked on Q&A information sessions for new students and potential applicants.

EASTBIO lead student representative on industrial engagement committee

- Worked on maintaining the network between industrial partners and PhD students.

PROGRAMMING SKILLS

Programming: Python, Julia, R, Octave

- Python Packages for Scientific Computing and Machine Learning: NumPy, Pandas, SciPy, NetworkX, Matplotlib, Seaborn, Scikit-Learn, PyTorch, PyTorch Geometric, Tensorflow, JAX
- Julia Packages: DifferentialEquations, DiffEqFlux, Optimization, Catalyst
- Shell Command line operations, Git, basic software testing and packaging

Deep Learning for Healthcare Specialization Certificate on Coursera

Deep Learning Specialization Certificate on Coursera

Machine Learning Specialization Certificate on Coursera

ATTENDED CONFERENCES & WORKSHOPS

Synthetic Biology UK 2025: Poster presentation

Nov 2025

12th BioProNET2 Annual Meeting: Poster presentation

Oct 2025

Computational Intelligence methods for Bioinformatics and Biostatistics 2025: Flash talk

Sep 2025

Bioprocessing Summit Europe 2025: Speaker

Mar 2025

Synthetic Biology UK 2024: Oral Communication

Nov 2024

Protein & Antibody Engineering Summit (PEGS) Europe 2024: Poster presentation

Nov 2024

Synthetic Biology for Health and Sustainability: Short talk

Oct 2024

EASTBIO Annual Symposium 2024: Speaker, Student representative and Co-Organizer

Jun 2024

ChemEngDayUK 2024: Flash talk and poster presentation

Apr 2024

EMBL Symposium AI and biology: Flash talk and poster presentation

Mar 2024

BBSRC Inter-DTP Module in Software Engineering

Feb 2024

Synthetic Biology UK 2023: Poster presentation	Nov 2023
ELLIS Summer School on Machine Learning for Healthcare and Biology	Jun 2023
EASTBIO Annual Symposium 2023: Student representative and Co-Organizer	Jun 2023
AI, Engineering Biology & Beyond - Turing Institute Workshop: Volunteer in logistics	Mar 2023
Host of EASTBIO Thematic Meeting “Computational tools for integrated Omics”	Feb 2023